Lab 6

# Case Study:
# A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min
Groups of 4

LO3–LO5: combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

# Open your handout

## Lab 6 handout

`Lab6_ssi_outbreak_case_study.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

# About this case

- The **genomes are real**: MRSA isolates from a published outbreak investigation
- The **hospitals, wards, patients and dates are fictional**, so the case feels like our setting

> The real story is revealed at the end.

THE SITUATION

# District Hospital A · Monday, 16 March 2026

The IPC nurse writes to you:

> *"Since early February we've had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it's bad luck. Theatre staff say it's the ward. The ward says it's the theatre.*

# The question

> *The director asks whether we should **close Ward B**. That would cancel about 40 operations a week. We got genomes back last week for all MRSA isolates since September.*
>
> ***Can you tell us what's going on by Wednesday's IPC meeting?"***

# Your evidence

1. **Microreact project** from Lab 5: tree + timeline + metadata
2. **SNP matrix:** `koser_snp_matrix.tsv` (opens in Excel)
3. **Pathogenwatch results** from Lab 5: ST and AMR
4. **Metadata:** `lab6_metadata.csv`

# Group roles

| Role | Job |
|---|---|
| **Genomics lead** | Reads the tree and SNP matrix |
| **Epidemiologist** | Builds the timeline: who was where, when |
| **Clinician / IPC** | Thinks through transmission routes and actions |
| **Communicator** | Writes and gives the 5-minute briefing |

# Plan for this lab

| Part | What | Time |
|---|---|---|
| A | Describe the cases | 15 min |
| B | What does the genome say? | 25 min |
| C | Genomics + epidemiology together | 20 min |
| D | Prepare the briefing | 25 min |
| E | Briefings to the IPC committee | 15 min |

# Describe the cases

Sketch an **epidemic curve**: weeks across, number of cases up.
Two colours: **Ward B** and **other wards / hospitals**.

**QUESTIONS**

1. How many MRSA isolates in total? How many from Ward B since 1 February?
2. Do all Ward B cases look the same on the **lab report**? Does that make them one outbreak?

# What does the genome say?

1. **Sequence types:** group the isolates by ST (Pathogenwatch)
2. **SNP distances:** highlight all pairs **≤ 30 SNPs** apart
3. Which isolates form a **cluster**, all close to each other?
4. **Tree:** colour by **ward**, then by **theatre**. Is the cluster a tight group on short branches?

≤ 30 SNPs: our working threshold for possible recent *S. aureus* transmission. Published thresholds vary.

# Put genomics and epidemiology together

For **each** Ward B case since February:

| Patient | In the cluster? | Ward / theatre / time link? | Linked · not linked · uncertain |
|---|---|---|---|
| | | | |
| | | | |

Use the full table in your handout.

PART C

# Discuss in your group

1. Is there an outbreak? How confident are you?
2. Is a Ward B patient **outside** the cluster, though on the ward at the same time?
3. Is a cluster patient **not** on Ward B? What do they share? → **likely route?**
4. Are isolates from **other hospitals** in the cluster?
5. What **other information** would you want?

# Prepare a 5-minute briefing

For the IPC committee: director, chief surgeon, nursing lead, lab head. **Not genomics experts.**

**RULES**

- One flip-chart page or 3 slides
- Explain any jargon: “SNP” → “genetic differences”
- **No more than one figure**

PART D

# Briefing template

1. **Bottom line:** “We found / did not find evidence of an outbreak of ___ linked to ____.”
2. **What we did:** “We sequenced __ MRSA isolates from __ to __.”
3. **What we found:** one visual + 2–3 bullets
4. **What it means:** which cases are linked; the likely route
5. **Recommendations:** 3 actions, with who and by when
6. **What we don’t know yet**

# Briefings to the IPC committee

Each group has **5 minutes**. The committee must ask at least one question:

**“Should we close Ward B?”**

**“Is it the surgeons' fault?”**

**“How sure are you?”**

**“What will this cost?”**

# Pause

Instructor: don’t go to the next slide until every group has given its briefing

# The reveal

The genomes come from **Köser et al., *NEJM* 2012**: a suspected MRSA outbreak in a **neonatal unit** in Cambridge, UK.

- Sequencing separated the **outbreak isolates** from unrelated MRSA that looked identical on routine tests
- A follow-up study (**Harris et al. 2013**) traced transmission into the community and found a **staff carrier**, which ended the outbreak

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.

# Take-home messages

1. **Same species + same antibiogram ≠ same outbreak**
2. Genomics finds **hidden links**: shared theatre, staff, equipment
3. Genomics can **prevent unnecessary actions**, like closing a ward
4. Genomics is only useful with **good metadata** and **clear communication**